

Manufacturing Digitalisation Framework

Thought Leadership Whitepaper | Industry 4.0 Practice

Abstract

Manufacturers across sectors are grappling with how to sequence their Industry 4.0 journey without over-investing in technology that outpaces organisational readiness. This whitepaper presents our Manufacturing Digitalisation Framework—a five-dimension maturity model that helps manufacturers assess their current state, prioritise investments, and build a connected factory architecture that delivers measurable returns at every stage.

1. The Industry 4.0 Maturity Model

We assess manufacturers across five dimensions, each scored on a 1–5 scale:

- Data Infrastructure: Connectivity from machine to cloud (PLC → Edge → Historian → Data Lake)
- Process Standardisation: Digital work instructions, standard operating procedures, quality plans
- Workforce Capability: Digital literacy of operators, supervisors, and technical staff
- Technology Readiness: ERP, MES, SCADA, and IoT platform modernity and integration readiness
- Leadership Commitment: Executive sponsorship quality and digital transformation governance

2. The Connected Factory Architecture

A Connected Factory is not a single product — it is an architecture of integrated layers:

- Level 0 — Field: Sensors, actuators, PLCs collecting raw process data
- Level 1 — Control: SCADA/DCS systems providing real-time process visibility
- Level 2 — Operations: MES layer connecting business intent to shop-floor execution
- Level 3 — Enterprise: ERP, supply chain, and quality systems driving business decisions
- Level 4 — Analytics: BI, ML, and digital twin providing predictive and prescriptive insights

3. Digital Twin in Manufacturing

A digital twin is a virtual replica of a physical asset, process, or system that updates in real time from IoT sensor data. In manufacturing, digital twins enable: process simulation to test changes before physical implementation, root-cause analysis using historical replay, and predictive what-if modelling for capacity planning and energy optimisation.

4. Key Adoption Statistics

- Global smart manufacturing market projected to reach USD 658 billion by 2030 (CAGR 14.5%)
- Manufacturing firms with mature digital infrastructure achieve 2.5× higher EBITDA growth
- Predictive maintenance adoption reduces unplanned downtime by 30–50% on average

- Only 29% of Indian manufacturers have reached Maturity Level 3 or above (2023 survey, n=340)
- ROI payback period for connected factory programmes: median 18–24 months

5. Recommended Transformation Roadmap

Regardless of starting point, we recommend a consistent sequencing: (1) Baseline with data infrastructure and connectivity; (2) Quick wins through OEE dashboards and anomaly alerting; (3) Optimise with ML-based predictive models; (4) Integrate across the value chain with supplier and customer digital interfaces; (5) Innovate with digital twins and autonomous operations. Skipping steps creates technical debt that is expensive to unwind.